# Manage network IP subnets

ONTAP 9.13.1 REST API reference

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# Manage network IP subnets

## Network IP subnets endpoint overview

### Overview

The following operations are supported:

- Creation: POST network/ip/subnets
- Collection Get: GET network/ip/subnets
- Instance Get: GET network/ip/subnets/{uuid}
- Instance Patch: PATCH network/ip/subnets/{uuid}
- Instance Delete: DELETE network/ip/subnets/{uuid}

### Retrieving IP subnet information

The IP subnets GET API retrieves and displays relevant information pertaining to the subnets configured in the cluster. The response can contain a list of multiple subnets or a specific subnet.

### Examples

#### Retrieving all subnets in the cluster

The following example shows the list of all subnets configured in a cluster.

```
# The API:
/api/network/ip/subnets

# The call:
curl -X GET "https://<mgmt-ip>/api/network/ip/subnets" -H "accept:
application/hal+json"

# The response:
{
"records": [
  {
    "uuid": "451d8d99-582c-11ec-8572-005056acd597",
    "name": "Subnet-002",
    "_links": {
      "self": {
        "href": "/api/network/ip/subnets/451d8d99-582c-11ec-8572-
005056acd597"
      }
    }
  },
  {
    "uuid": "615b722f-5795-11ec-8572-005056acd597",
    "name": "Subnet-001",
    "_links": {
      "self": {
        "href": "/api/network/ip/subnets/615b722f-5795-11ec-8572-
005056acd597"
      }
    }
  }
],
"num_records": 2,
"_links": {
  "self": {
    "href": "/api/network/ip/subnets"
  }
}
}
```

#### Retrieving a specific subnet

The following example shows the response when a specific subnet is requested. This is equivalent to fields=*, which returns most of the fields. The system returns an error when there is no subnet with the requested UUID.

```
# The API:
/api/network/ip/subnets/{uuid}

# The call:
curl -X GET "https://<mgmt-ip>/api/network/ip/subnets/451d8d99-582c-11ec-
8572-005056acd597" -H "accept: application/hal+json"

# The response:
{
"uuid": "451d8d99-582c-11ec-8572-005056acd597",
"name": "Subnet-002",
"ipspace": {
  "uuid": "6f62c691-5780-11ec-8572-005056acd597",
  "name": "Default",
  "_links": {
    "self": {
      "href": "/api/network/ipspaces/6f62c691-5780-11ec-8572-005056acd597"
    }
  }
},
"broadcast_domain": {
  "uuid": "9a1dce3b-5780-11ec-8572-005056acd597",
  "name": "Default",
  "_links": {
    "self": {
      "href": "/api/network/ethernet/broadcast-domains/9a1dce3b-5780-11ec-
8572-005056acd597"
    }
  }
},
"subnet": {
  "address": "10.2.1.0",
  "netmask": "24",
  "family": "ipv4"
},
"gateway": "10.2.1.1",
"_links": {
  "self": {
    "href": "/api/network/ip/subnets/451d8d99-582c-11ec-8572-005056acd597"
  }
}
}
```

#### Retrieving all the fields for a specific subnet

The following example shows the response when all the fields for a specific subnet are requested, returning everything that fields=* returns plus the IP ranges and count fields. The system returns an error when there is no subnet with the requested UUID.

```
# The API:
/api/network/ip/subnets/{uuid}

# The call:
curl -X GET "https://<mgmt-ip>/api/network/ip/subnets/451d8d99-582c-11ec-
8572-005056acd597?fields=**" -H "accept: application/hal+json"

# The response:
{
"uuid": "451d8d99-582c-11ec-8572-005056acd597",
"name": "Subnet-002",
"ipspace": {
  "uuid": "6f62c691-5780-11ec-8572-005056acd597",
  "name": "Default",
  "_links": {
    "self": {
      "href": "/api/network/ipspaces/6f62c691-5780-11ec-8572-005056acd597"
    }
  }
},
"broadcast_domain": {
  "uuid": "9a1dce3b-5780-11ec-8572-005056acd597",
  "name": "Default",
  "_links": {
    "self": {
      "href": "/api/network/ethernet/broadcast-domains/9a1dce3b-5780-11ec-
8572-005056acd597"
    }
  }
},
"subnet": {
  "address": "10.2.1.0",
  "netmask": "24",
  "family": "ipv4"
},
"gateway": "10.2.1.1",
"ip_ranges": [
  {
    "start": "10.2.1.10",
```

```
    "end": "10.2.1.22",
    "family": "ipv4"
  },
  {
    "start": "10.2.1.101",
    "end": "10.2.1.200",
    "family": "ipv4"
  }
],
"available_ip_ranges": [
  {
    "start": "10.2.1.10",
    "end": "10.2.1.22",
    "family": "ipv4"
  },
  {
    "start": "10.2.1.101",
    "end": "10.2.1.200",
    "family": "ipv4"
  }
],
"total_count": 113,
"used_count": 0,
"available_count": 113,
"_links": {
  "self": {
    "href": "/api/network/ip/subnets/451d8d99-582c-11ec-8572-
005056acd597?fields=**"
  }
}
}
```


## Creating IP subnets

You can use the IP subnets POST API to create IP subnets as shown in the following examples.

## Examples

**Creating an IP subnet using the minimum number of parameters.**

The following example shows the record returned after the creation of an IP subnet.

```
# The API:
/api/network/ip/subnets

# The call:
curl -X POST "https://<mgmt-ip>/api/network/ip/subnets" -H "accept:
application/hal+json" -d '{ "name": "Subnet-003", "broadcast_domain": {
"uuid": "6577524b-5863-11ec-8981-005056a7077f" }, "subnet": { "address":
"10.3.0.0", "netmask": "16" } }'

# The response:
{
}

# The result:
curl -X GET "https://<mgmt-ip>/api/network/ip/subnets?name=Subnet-
003&fields=**" -H "accept: application/hal+json"
{
"records": [
  {
    "uuid": "79ff5a5e-59b7-11ec-8981-005056a7077f",
    "name": "Subnet-003",
    "ipspace": {
      "uuid": "36569d0f-5863-11ec-8981-005056a7077f",
      "name": "Default",
      "_links": {
        "self": {
          "href": "/api/network/ipspaces/36569d0f-5863-11ec-8981-
005056a7077f"
        }
      }
    },
    "broadcast_domain": {
      "uuid": "6577524b-5863-11ec-8981-005056a7077f",
      "name": "Default",
      "_links": {
        "self": {
          "href": "/api/network/ethernet/broadcast-domains/6577524b-5863-
11ec-8981-005056a7077f"
        }
      }
    },
    "subnet": {
      "address": "10.3.0.0",
      "netmask": "16",
      "family": "ipv4"
    },
```

```
    "total_count": 0,
    "used_count": 0,
    "available_count": 0,
    "_links": {
      "self": {
        "href": "/api/network/ip/subnets/79ff5a5e-59b7-11ec-8981-
005056a7077f?fields=**"
      }
    }
  }
],
"num_records": 1,
"_links": {
  "self": {
    "href": "/api/network/ip/subnets?name=Subnet-003&fields=**"
  }
}
}
```

**Creating an IP subnet using all parameters.**

The following example shows the record returned after the creation of an IP subnet setting all parameters.

```
# The API:
/api/network/ip/subnets

# The call:
curl -X POST "https://<mgmt-ip>/api/network/ip/subnets" -H "accept:
application/hal+json" -d '{ "name": "Subnet-004", "ipspace": { "name":
"Default", "uuid": "36569d0f-5863-11ec-8981-005056a7077f" },
"broadcast_domain": { "name": "Default", "uuid": "6577524b-5863-11ec-8981-
005056a7077f" }, "subnet": { "address": "10.4.1.0", "netmask": "24" },
"gateway": "10.4.1.1", "ip_ranges": [ { "start": "10.4.1.30", "end":
"10.4.1.39" }, { "start": "10.4.1.150", "end": "10.4.1.229" } ],
"fail_if_lifs_conflict": "false" }'

# The response:
{
}

# The result:
curl -X GET "https://<mgmt-ip>/api/network/ip/subnets?name=Subnet-
```

```
004&fields=**" -H "accept: application/hal+json"
{
"records": [
  {
    "uuid": "0e0a19e7-59ba-11ec-8981-005056a7077f",
    "name": "Subnet-004",
    "ipspace": {
      "uuid": "36569d0f-5863-11ec-8981-005056a7077f",
      "name": "Default",
      "_links": {
        "self": {
          "href": "/api/network/ipspaces/36569d0f-5863-11ec-8981-
005056a7077f"
        }
      }
    },
    "broadcast_domain": {
      "uuid": "6577524b-5863-11ec-8981-005056a7077f",
      "name": "Default",
      "_links": {
        "self": {
          "href": "/api/network/ethernet/broadcast-domains/6577524b-5863-
11ec-8981-005056a7077f"
        }
      }
    },
    "subnet": {
      "address": "10.4.1.0",
      "netmask": "24",
      "family": "ipv4"
    },
    "gateway": "10.4.1.1",
    "ip_ranges": [
      {
        "start": "10.4.1.30",
        "end": "10.4.1.39",
        "family": "ipv4"
      },
      {
        "start": "10.4.1.150",
        "end": "10.4.1.229",
        "family": "ipv4"
      }
    ],
    "available_ip_ranges": [
      {
```

```
        "start": "10.4.1.30",
        "end": "10.4.1.39",
        "family": "ipv4"
      },
      {
        "start": "10.4.1.150",
        "end": "10.4.1.229",
        "family": "ipv4"
      }
    ],
    "total_count": 90,
    "used_count": 0,
    "available_count": 90,
    "_links": {
      "self": {
        "href": "/api/network/ip/subnets/0e0a19e7-59ba-11ec-8981-
005056a7077f?fields=**"
      }
    }
  }
],
"num_records": 1,
"_links": {
  "self": {
    "href": "/api/network/ip/subnets?name=Subnet-004&fields=**"
  }
}
}
```


## Updating IP subnets

You can use the IP subnets PATCH API to update the attributes of an IP subnet.

## Examples

### Updating the name of an IP subnet

The following example shows how the PATCH request changes the name.

```
# The API:
/api/network/ip/subnets/{uuid}

# The call:
curl -X PATCH "https://<mgmt-ip>/api/network/ip/subnets/0e0a19e7-59ba-
11ec-8981-005056a7077f" -H "accept: application/hal+json" -d '{ "name":
"Subnet-004-NewName" }'

# The response:
{
}

# The result:
curl -X GET "https://<mgmt-ip>/api/network/ip/subnets/0e0a19e7-59ba-11ec-
8981-005056a7077f?fields=name" -H "accept: application/hal+json"
{
"uuid": "0e0a19e7-59ba-11ec-8981-005056a7077f",
"name": "Subnet-004-NewName",
"_links": {
  "self": {
    "href": "/api/network/ip/subnets/0e0a19e7-59ba-11ec-8981-005056a7077f"
  }
}
}
```

#### Updating the ip_ranges of an IP subnet

The following example shows how the PATCH request updates the ip_ranges.

```
# The API:
/api/network/ip/subnets/{uuid}

# The call:
curl -X PATCH "https://<mgmt-ip>/api/network/ip/subnets/0e0a19e7-59ba-
11ec-8981-005056a7077f" -H "accept: application/hal+json" -d '{
"ip_ranges": [ { "start": "10.4.1.20", "end": "10.4.1.239" } ] }'

# The response:
{
}

# The result:
curl -X GET "https://<mgmt-ip>/api/network/ip/subnets/0e0a19e7-59ba-11ec-
8981-005056a7077f?fields=ip_ranges" -H "accept: application/hal+json"
{
"uuid": "0e0a19e7-59ba-11ec-8981-005056a7077f",
"name": "Subnet-004-NewName",
"ip_ranges": [
  {
    "start": "10.4.1.20",
    "end": "10.4.1.239",
    "family": "ipv4"
  }
],
"_links": {
  "self": {
    "href": "/api/network/ip/subnets/0e0a19e7-59ba-11ec-8981-005056a7077f"
  }
}
}
```

## Deleting IP subnets

You can use the IP subnets DELETE API to delete an IP subnet.

## Example

### Deleting an IP subnet

The following DELETE request deletes a specific network IP subnet.

```
# The API:
/api/network/ip/subnets/{uuid}

# The call:
curl -X DELETE "https://<mgmt-ip>/api/network/ip/subnets/0e0a19e7-59ba-
11ec-8981-005056a7077f"

# The response:
{
}

# The result:
curl -X GET "https://<mgmt-ip>/api/network/ip/subnets/0e0a19e7-59ba-11ec-
8981-005056a7077f" -H "accept: application/hal+json"
{
"error": {
  "message": "entry doesn't exist",
  "code": "4",
  "target": "uuid"
}
}
```

# Retrieve details for all subnets

GET `/network/ip/subnets`

**Introduced In:** 9.11

Retrieves details for all subnets.

## Related ONTAP Commands

- `network subnet show`

## Parameters

| Name | Type | In | Required | Description |
|---|---|---|---|---|
| used_count | integer | query | False | Filter by used_count |
| broadcast_domain.name | string | query | False | Filter by broadcast_domain.name |

| Name | Type | In | Required | Description |
| --- | --- | --- | --- | --- |
| broadcast_domain.uuid | string | query | False | Filter by broadcast_domain.uuid |
| name | string | query | False | Filter by name |
| uuid | string | query | False | Filter by uuid |
| gateway | string | query | False | Filter by gateway |
| subnet.netmask | string | query | False | Filter by subnet.netmask |
| subnet.address | string | query | False | Filter by subnet.address |
| subnet.family | string | query | False | Filter by subnet.family |
| available_ip_ranges.start | string | query | False | Filter by available_ip_ranges.start |
| available_ip_ranges.end | string | query | False | Filter by available_ip_ranges.end |
| available_ip_ranges.family | string | query | False | Filter by available_ip_ranges.family |
| ip_ranges.start | string | query | False | Filter by ip_ranges.start |
| ip_ranges.end | string | query | False | Filter by ip_ranges.end |
| ip_ranges.family | string | query | False | Filter by ip_ranges.family |
| ipspace.name | string | query | False | Filter by ipspace.name |
| ipspace.uuid | string | query | False | Filter by ipspace.uuid |

| Name | Type | In | Required | Description |
|---|---|---|---|---|
| total_count | integer | query | False | Filter by total_count |
| available_count | integer | query | False | Filter by available_count |
| fields | array[string] | query | False | Specify the fields to return. |
| max_records | integer | query | False | Limit the number of records returned. |
| return_records | boolean | query | False | The default is true for GET calls. When set to false, only the number of records is returned.<br>• Default value: 1 |
| return_timeout | integer | query | False | The number of seconds to allow the call to execute before returning. When iterating over a collection, the default is 15 seconds. ONTAP returns earlier if either max records or the end of the collection is reached.<br>• Default value: 1<br>• Max value: 120<br>• Min value: 0 |
| order_by | array[string] | query | False | Order results by specified fields and optional [asc |

## Response

```
Status: 200, Ok
```

| Name | Type | Description |
| --- | --- | --- |
| _links | _links | |
| num_records | integer | Number of records |
| records | array[ip_subnet] | |

**Example response**


```
{
  "_links": {
    "next": {
      "href": "/api/resourcelink"
    },
    "self": {
      "href": "/api/resourcelink"
    }
  },
  "num_records": 1,
  "records": {
    "_links": {
      "self": {
        "href": "/api/resourcelink"
      }
    },
    "available_count": 0,
    "available_ip_ranges": {
      "end": "10.10.10.7",
      "family": "ipv4",
      "start": "10.10.10.7"
    },
    "broadcast_domain": {
      "_links": {
        "self": {
          "href": "/api/resourcelink"
        }
      },
      "name": "bd1",
      "uuid": "1cd8a442-86d1-11e0-ae1c-123478563412"
    },
    "gateway": "10.1.1.1",
    "ip_ranges": {
      "end": "10.10.10.7",
      "family": "ipv4",
      "start": "10.10.10.7"
    },
    "ipspace": {
      "_links": {
        "self": {
          "href": "/api/resourcelink"
        }
      },
      "name": "exchange",
```

```
      "uuid": "1cd8a442-86d1-11e0-ae1c-123478563412"
    },
    "name": "subnet1",
    "subnet": {
      "address": "10.10.10.7",
      "family": "ipv4",
      "netmask": "24"
    },
    "total_count": 0,
    "used_count": 0,
    "uuid": "1cd8a442-86d1-11e0-ae1c-123478563412"
  }
}
```


## Error

```
Status: Default, Error
```

| Name | Type | Description |
|---|---|---|
| error | error | |

### Example error


```
{
  "error": {
    "arguments": {
      "code": "string",
      "message": "string"
    },
    "code": "4",
    "message": "entry doesn't exist",
    "target": "uuid"
  }
}
```


## Definitions

**See Definitions**

href

| Name | Type | Description |
|---|---|---|
| href | string | |

_links

| Name | Type | Description |
|---|---|---|
| next | href | |
| self | href | |

_links

| Name | Type | Description |
|---|---|---|
| self | href | |

ip_address_range

IP address range

| Name | Type | Description |
|---|---|---|
| end | string | IPv4 or IPv6 address |
| family | string | IPv4 or IPv6 |
| start | string | IPv4 or IPv6 address |

broadcast_domain

The broadcast domain that the subnet is associated with. Either the UUID or name must be supplied on POST.

| Name | Type | Description |
|---|---|---|
| _links | _links | |
| name | string | Name of the broadcast domain, scoped to its IPspace |
| uuid | string | Broadcast domain UUID |

ipspace

The IPspace that the subnet is associated with. Either the UUID or name must be supplied on POST.

| Name | Type | Description |
| --- | --- | --- |
| _links | _links | |
| name | string | IPspace name |
| uuid | string | IPspace UUID |

ip_info

IP information

| Name | Type | Description |
| --- | --- | --- |
| address | string | IPv4 or IPv6 address |
| family | string | IPv4 or IPv6 |
| netmask | string | Input as netmask length (16) or IPv4 mask (255.255.0.0). For IPv6, the default value is 64 with a valid range of 1 to 127. Output is always netmask length. |

ip_subnet

| Name | Type | Description |
| --- | --- | --- |
| _links | _links | |
| available_count | integer | |
| available_ip_ranges | array[ip_address_range] | |
| broadcast_domain | broadcast_domain | The broadcast domain that the subnet is associated with. Either the UUID or name must be supplied on POST. |
| fail_if_lifs_conflict | boolean | This action will fail if any existing interface is using an IP address in the ranges provided. Set this to false to associate any manually addressed interfaces with the subnet and allow the action to succeed. |
| gateway | string | The IP address of the gateway for this subnet. |
| ip_ranges | array[ip_address_range] | |

| Name | Type | Description |
| --- | --- | --- |
| ipspace | ipspace | The IPspace that the subnet is associated with. Either the UUID or name must be supplied on POST. |
| name | string | Subnet name |
| subnet | ip_info | IP information |
| total_count | integer | |
| used_count | integer | |
| uuid | string | The UUID that uniquely identifies the subnet. |

error_arguments

| Name | Type | Description |
| --- | --- | --- |
| code | string | Argument code |
| message | string | Message argument |

error

| Name | Type | Description |
| --- | --- | --- |
| arguments | array[error_arguments] | Message arguments |
| code | string | Error code |
| message | string | Error message |
| target | string | The target parameter that caused the error. |

# Create a new named subnet

POST `/network/ip/subnets`

**Introduced In:** 9.11

Creates a new named subnet.

## Required properties

- `name` - Name of the subnet to create.
- `broadcast_domain` - Broadcast domain containing the subnet.
- `ipspace` - IPspace containing the subnet. Required only if `broadcast_domain.uuid` is not provided.
- `subnet.address` - IP address for the subnet.
- `subnet.netmask` - IP netmask of the subnet.

## Recommended property values

## Default property values

If not specified in POST, the following default property values are assigned:

- `gateway` - no gateway
- `ip_ranges` - empty
- `fail_if_lifs_conflict` - *true*

## Related ONTAP commands

- `network subnet create`

## Parameters

| Name | Type | In | Required | Description |
|---|---|---|---|---|
| return_records | boolean | query | False | The default is false. If set to true, the records are returned.<br>• Default value: |

## Request Body

| Name | Type | Description |
|---|---|---|
| _links | _links | |
| available_count | integer | |
| available_ip_ranges | array[ip_address_range] | |
| broadcast_domain | broadcast_domain | The broadcast domain that the subnet is associated with. Either the UUID or name must be supplied on POST. |

| Name | Type | Description |
| --- | --- | --- |
| fail_if_lifs_conflict | boolean | This action will fail if any existing interface is using an IP address in the ranges provided. Set this to false to associate any manually addressed interfaces with the subnet and allow the action to succeed. |
| gateway | string | The IP address of the gateway for this subnet. |
| ip_ranges | array[ip_address_range] | |
| ipspace | ipspace | The IPspace that the subnet is associated with. Either the UUID or name must be supplied on POST. |
| name | string | Subnet name |
| subnet | ip_info | IP information |
| total_count | integer | |
| used_count | integer | |
| uuid | string | The UUID that uniquely identifies the subnet. |

**Example request**


```
{
  "_links": {
    "self": {
      "href": "/api/resourcelink"
    }
  },
  "available_count": 0,
  "available_ip_ranges": {
    "end": "10.10.10.7",
    "family": "ipv4",
    "start": "10.10.10.7"
  },
  "broadcast_domain": {
    "_links": {
      "self": {
        "href": "/api/resourcelink"
      }
    },
    "name": "bd1",
    "uuid": "1cd8a442-86d1-11e0-ae1c-123478563412"
  },
  "gateway": "10.1.1.1",
  "ip_ranges": {
    "end": "10.10.10.7",
    "family": "ipv4",
    "start": "10.10.10.7"
  },
  "ipspace": {
    "_links": {
      "self": {
        "href": "/api/resourcelink"
      }
    },
    "name": "exchange",
    "uuid": "1cd8a442-86d1-11e0-ae1c-123478563412"
  },
  "name": "subnet1",
  "subnet": {
    "address": "10.10.10.7",
    "family": "ipv4",
    "netmask": "24"
  },
  "total_count": 0,
  "used_count": 0,
```

```
  "uuid": "1cd8a442-86d1-11e0-ae1c-123478563412"
}
```

## Response

```
Status: 201, Created
```

### Headers

| Name | Description | Type |
|---|---|---|
| Location | Useful for tracking the resource location | string |

## Error

```
Status: Default
```

ONTAP Error Response Codes

| Error Code | Description |
|---|---|
| 1377660 | A subnet with the name already exists in the IPspace. |
| 1377661 | Subnet in IPspace cannot use subnet address because that address is already used by another subnet in the same IPspace. |
| 1377662 | The IP range address is not within the subnet in IPspace. |
| 1377663 | The specified IP address range of subnet in IPspace contains an address already in use by a LIF. |
| 1377664 | The specified IP address range of subnet in IPspace contains an address already in use by the Service Processor. |
| 1377673 | The addresses provided must have the same address family. |
| 1377681 | Cannot update LIF associations for LIF. The broadcast domain of the LIF does not match the broadcast domain of the subnet. |
| 1966269 | IPv4 Addresses must have a prefix length between 1 and 32. |
| 1967082 | The specified ipspace.name does not match the IPspace name of specified ipspace.uuid |

| Error Code | Description |
| --- | --- |
| 53282568 | The subnet.address must be specified together with subnet.netmask. |
| 53282569 | The specified subnet.netmask is not valid. |
| 53282570 | Each pair of ranges must have ip_ranges.start less than or equal to ip_ranges.end. |
| 53282571 | The ip_ranges.start and ip_ranges.end fields must have the same number of items. |
| 53282573 | Broadcast domain is a required parameter. The broadcast_domain.name and/or the broadcast_domain.uuid must be specified. |
| 53282574 | The specified broadcast_domain and ipspace parameters do not match. |
| 53282575 | Operation might have left configuration in an inconsistent state. Unable to set UUID for created entry. |
| 53282576 | The specified ipspace.uuid is invalid. |
| 53282577 | The specified broadcast_domain.uuid is invalid. |
| 53282578 | The specified broadcast_domain.name does not match the IPspace name of specified broadcast_domain.uuid |
| 53282579 | Missing the ipspace.name or ipspace.uuid parameter. |

| Name | Type | Description |
| --- | --- | --- |
| error | error | |

**Example error**

```
{
  "error": {
    "arguments": {
      "code": "string",
      "message": "string"
    },
    "code": "4",
    "message": "entry doesn't exist",
    "target": "uuid"
  }
}
```

## Definitions

**See Definitions**

href

| Name | Type | Description |
|---|---|---|
| href | string | |

_links

| Name | Type | Description |
|---|---|---|
| self | href | |

ip_address_range

IP address range

| Name | Type | Description |
|---|---|---|
| end | string | IPv4 or IPv6 address |
| family | string | IPv4 or IPv6 |
| start | string | IPv4 or IPv6 address |

broadcast_domain

The broadcast domain that the subnet is associated with. Either the UUID or name must be supplied on POST.

| Name | Type | Description |
|---|---|---|
| _links | _links | |
| name | string | Name of the broadcast domain, scoped to its IPspace |
| uuid | string | Broadcast domain UUID |

ipspace

The IPspace that the subnet is associated with. Either the UUID or name must be supplied on POST.

| Name | Type | Description |
|---|---|---|
| _links | _links | |
| name | string | IPspace name |
| uuid | string | IPspace UUID |

ip_info

IP information

| Name | Type | Description |
|---|---|---|
| address | string | IPv4 or IPv6 address |
| family | string | IPv4 or IPv6 |
| netmask | string | Input as netmask length (16) or IPv4 mask (255.255.0.0). For IPv6, the default value is 64 with a valid range of 1 to 127. Output is always netmask length. |

ip_subnet

| Name | Type | Description |
|---|---|---|
| _links | _links | |
| available_count | integer | |
| available_ip_ranges | array[ip_address_range] | |
| broadcast_domain | broadcast_domain | The broadcast domain that the subnet is associated with. Either the UUID or name must be supplied on POST. |
| fail_if_lifs_conflict | boolean | This action will fail if any existing interface is using an IP address in the ranges provided. Set this to false to associate any manually addressed interfaces with the subnet and allow the action to succeed. |
| gateway | string | The IP address of the gateway for this subnet. |
| ip_ranges | array[ip_address_range] | |
| ipspace | ipspace | The IPspace that the subnet is associated with. Either the UUID or name must be supplied on POST. |
| name | string | Subnet name |
| subnet | ip_info | IP information |

| Name | Type | Description |
|---|---|---|
| total_count | integer | |
| used_count | integer | |
| uuid | string | The UUID that uniquely identifies the subnet. |

error_arguments

| Name | Type | Description |
|---|---|---|
| code | string | Argument code |
| message | string | Message argument |

error

| Name | Type | Description |
|---|---|---|
| arguments | array[error_arguments] | Message arguments |
| code | string | Error code |
| message | string | Error message |
| target | string | The target parameter that caused the error. |

# Delete an IP subnet

DELETE `/network/ip/subnets/{uuid}`

**Introduced In:** 9.11

Deletes an IP subnet.

## Related ONTAP commands

- `network subnet delete`

## Parameters

| Name | Type | In | Required | Description |
|---|---|---|---|---|
| uuid | string | path | True | IP subnet UUID |

### Response

```
Status: 200, Ok
```

# Retrieve IP subnet details

GET `/network/ip/subnets/{uuid}`

**Introduced In:** 9.11

Retrieves details for a specific IP subnet.

### Related ONTAP commands

- `network subnet show`

### Parameters

| Name | Type | In | Required | Description |
|---|---|---|---|---|
| uuid | string | path | True | IP subnet UUID |
| fields | array[string] | query | False | Specify the fields to return. |

### Response

```
Status: 200, Ok
```

| Name | Type | Description |
|---|---|---|
| _links | _links | |
| available_count | integer | |
| available_ip_ranges | array[ip_address_range] | |
| broadcast_domain | broadcast_domain | The broadcast domain that the subnet is associated with. Either the UUID or name must be supplied on POST. |

| Name | Type | Description |
| --- | --- | --- |
| fail_if_lifs_conflict | boolean | This action will fail if any existing interface is using an IP address in the ranges provided. Set this to false to associate any manually addressed interfaces with the subnet and allow the action to succeed. |
| gateway | string | The IP address of the gateway for this subnet. |
| ip_ranges | array[ip_address_range] | |
| ipspace | ipspace | The IPspace that the subnet is associated with. Either the UUID or name must be supplied on POST. |
| name | string | Subnet name |
| subnet | ip_info | IP information |
| total_count | integer | |
| used_count | integer | |
| uuid | string | The UUID that uniquely identifies the subnet. |

**Example response**


```
{
  "_links": {
    "self": {
      "href": "/api/resourcelink"
    }
  },
  "available_count": 0,
  "available_ip_ranges": {
    "end": "10.10.10.7",
    "family": "ipv4",
    "start": "10.10.10.7"
  },
  "broadcast_domain": {
    "_links": {
      "self": {
        "href": "/api/resourcelink"
      }
    },
    "name": "bd1",
    "uuid": "1cd8a442-86d1-11e0-ae1c-123478563412"
  },
  "gateway": "10.1.1.1",
  "ip_ranges": {
    "end": "10.10.10.7",
    "family": "ipv4",
    "start": "10.10.10.7"
  },
  "ipspace": {
    "_links": {
      "self": {
        "href": "/api/resourcelink"
      }
    },
    "name": "exchange",
    "uuid": "1cd8a442-86d1-11e0-ae1c-123478563412"
  },
  "name": "subnet1",
  "subnet": {
    "address": "10.10.10.7",
    "family": "ipv4",
    "netmask": "24"
  },
  "total_count": 0,
  "used_count": 0,
```

```
  "uuid": "1cd8a442-86d1-11e0-ae1c-123478563412"
}
```

## Error

```
Status: Default, Error
```

| Name | Type | Description |
|---|---|---|
| error | error | |

### Example error

```
{
  "error": {
    "arguments": {
      "code": "string",
      "message": "string"
    },
    "code": "4",
    "message": "entry doesn't exist",
    "target": "uuid"
  }
}
```

## Definitions

**See Definitions**

href

| Name | Type | Description |
|---|---|---|
| href | string | |

_links

| Name | Type | Description |
|---|---|---|
| self | href | |

ip_address_range

IP address range

| Name | Type | Description |
|---|---|---|
| end | string | IPv4 or IPv6 address |
| family | string | IPv4 or IPv6 |
| start | string | IPv4 or IPv6 address |

broadcast_domain

The broadcast domain that the subnet is associated with. Either the UUID or name must be supplied on POST.

| Name | Type | Description |
|---|---|---|
| _links | _links | |
| name | string | Name of the broadcast domain, scoped to its IPspace |
| uuid | string | Broadcast domain UUID |

ipspace

The IPspace that the subnet is associated with. Either the UUID or name must be supplied on POST.

| Name | Type | Description |
|---|---|---|
| _links | _links | |
| name | string | IPspace name |
| uuid | string | IPspace UUID |

ip_info

IP information

| Name | Type | Description |
|---|---|---|
| address | string | IPv4 or IPv6 address |
| family | string | IPv4 or IPv6 |
| netmask | string | Input as netmask length (16) or IPv4 mask (255.255.0.0). For IPv6, the default value is 64 with a valid range of 1 to 127. Output is always netmask length. |

error_arguments

| Name | Type | Description |
|---|---|---|
| code | string | Argument code |
| message | string | Message argument |

error

| Name | Type | Description |
|---|---|---|
| arguments | array[error_arguments] | Message arguments |
| code | string | Error code |
| message | string | Error message |
| target | string | The target parameter that caused the error. |

# Update an IP subnet

PATCH `/network/ip/subnets/{uuid}`

**Introduced In:** 9.11

Updates an IP subnet.

## Related ONTAP commands

- `network subnet modify`
- `network subnet rename`
- `network subnet add-ranges`
- `network subnet remove-ranges`

## Parameters

| Name | Type | In | Required | Description |
|---|---|---|---|---|
| uuid | string | path | True | IP subnet UUID |

## Request Body

| Name | Type | Description |
|---|---|---|
| _links | _links | |
| available_count | integer | |
| available_ip_ranges | array[ip_address_range] | |
| broadcast_domain | broadcast_domain | The broadcast domain that the subnet is associated with. Either the UUID or name must be supplied on POST. |
| fail_if_lifs_conflict | boolean | This action will fail if any existing interface is using an IP address in the ranges provided. Set this to false to associate any manually addressed interfaces with the subnet and allow the action to succeed. |
| gateway | string | The IP address of the gateway for this subnet. |
| ip_ranges | array[ip_address_range] | |
| ipspace | ipspace | The IPspace that the subnet is associated with. Either the UUID or name must be supplied on POST. |
| name | string | Subnet name |
| subnet | ip_info | IP information |
| total_count | integer | |

| Name | Type | Description |
| --- | --- | --- |
| used_count | integer | |
| uuid | string | The UUID that uniquely identifies the subnet. |

**Example request**


```
{
  "_links": {
    "self": {
      "href": "/api/resourcelink"
    }
  },
  "available_count": 0,
  "available_ip_ranges": {
    "end": "10.10.10.7",
    "family": "ipv4",
    "start": "10.10.10.7"
  },
  "broadcast_domain": {
    "_links": {
      "self": {
        "href": "/api/resourcelink"
      }
    },
    "name": "bd1",
    "uuid": "1cd8a442-86d1-11e0-ae1c-123478563412"
  },
  "gateway": "10.1.1.1",
  "ip_ranges": {
    "end": "10.10.10.7",
    "family": "ipv4",
    "start": "10.10.10.7"
  },
  "ipspace": {
    "_links": {
      "self": {
        "href": "/api/resourcelink"
      }
    },
    "name": "exchange",
    "uuid": "1cd8a442-86d1-11e0-ae1c-123478563412"
  },
  "name": "subnet1",
  "subnet": {
    "address": "10.10.10.7",
    "family": "ipv4",
    "netmask": "24"
  },
  "total_count": 0,
  "used_count": 0,
```

```
  "uuid": "1cd8a442-86d1-11e0-ae1c-123478563412"
}
```

## Response

```
Status: 200, Ok
```

## Error

```
Status: Default
```

Fill error codes below. ONTAP Error Response Codes

| Error Code | Description |
|---|---|
| 1377658 | Invalid gateway for subnet in IPspace. |
| 1377659 | Subnet would overlap with existing subnet named in IPspace. |
| 1377660 | A subnet with the name already exists in the IPspace. |
| 1377661 | Subnet in IPspace cannot use subnet address because that address is already used by subnet in the same IPspace. |
| 1377662 | The IP range address is not within the subnet in IPspace. |
| 1377663 | The specified IP address range of subnet in IPspace contains an address already in use by a LIF. |
| 1377664 | The specified IP address range of subnet in IPspace contains an address already in use by the Service Processor. |
| 1377669 | The specified gateway address exists in a configured range. |
| 1377673 | The addresses provided must have the same address family. |
| 1377674 | Modifying the netmask is not supported because the subnet has one or more interfaces associated with it. |
| 1377675 | The netmask of the interface did not match the netmask of the subnet. |
| 1377681 | Cannot update LIF associations for LIF. The broadcast domain of the LIF does not match the broadcast domain of the subnet. |

| Error Code | Description |
| --- | --- |
| 1966269 | IPv4 Addresses must have a prefix length between 1 and 32. |
| 1966778 | IPv6 addresses must have a prefix length between 1 and 127. |
| 53282568 | The subnet.address must be specified together with subnet.netmask. |
| 53282569 | The specified subnet.netmask is not valid. |
| 53282570 | Each pair of ranges must have ip_ranges.start less than or equal to ip_ranges.end. |
| 53282571 | The ip_ranges.start and ip_ranges.end fields must have the same number of items. |
| 53282572 | PATCH partially succeeded with error. |

## Definitions

**See Definitions**

href

| Name | Type | Description |
|---|---|---|
| href | string | |

_links

| Name | Type | Description |
|---|---|---|
| self | href | |

ip_address_range

IP address range

| Name | Type | Description |
|---|---|---|
| end | string | IPv4 or IPv6 address |
| family | string | IPv4 or IPv6 |
| start | string | IPv4 or IPv6 address |

broadcast_domain

The broadcast domain that the subnet is associated with. Either the UUID or name must be supplied on POST.

| Name | Type | Description |
|---|---|---|
| _links | _links | |
| name | string | Name of the broadcast domain, scoped to its IPspace |
| uuid | string | Broadcast domain UUID |

ipspace

The IPspace that the subnet is associated with. Either the UUID or name must be supplied on POST.

| Name | Type | Description |
|---|---|---|
| _links | _links | |
| name | string | IPspace name |
| uuid | string | IPspace UUID |

ip_info

IP information

| Name | Type | Description |
|---|---|---|
| address | string | IPv4 or IPv6 address |
| family | string | IPv4 or IPv6 |
| netmask | string | Input as netmask length (16) or IPv4 mask (255.255.0.0). For IPv6, the default value is 64 with a valid range of 1 to 127. Output is always netmask length. |

ip_subnet

| Name | Type | Description |
|---|---|---|
| _links | _links | |
| available_count | integer | |
| available_ip_ranges | array[ip_address_range] | |
| broadcast_domain | broadcast_domain | The broadcast domain that the subnet is associated with. Either the UUID or name must be supplied on POST. |
| fail_if_lifs_conflict | boolean | This action will fail if any existing interface is using an IP address in the ranges provided. Set this to false to associate any manually addressed interfaces with the subnet and allow the action to succeed. |
| gateway | string | The IP address of the gateway for this subnet. |
| ip_ranges | array[ip_address_range] | |
| ipspace | ipspace | The IPspace that the subnet is associated with. Either the UUID or name must be supplied on POST. |
| name | string | Subnet name |
| subnet | ip_info | IP information |

| Name | Type | Description |
| --- | --- | --- |
| total_count | integer | |
| used_count | integer | |
| uuid | string | The UUID that uniquely identifies the subnet. |

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